

Credit Risk Report

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1. Summary

We developed a logistic-regression-based PD scorecard using 5,960 home-equity loans. The dataset was split into stratified training and validation samples. Predictors were assessed using univariate logistic regression and WoE/IV analysis, followed by variable transformation and binning. Several logistic regression specifications were developed and compared, with a reduced model selected based on statistical significance, likelihood-ratio testing, model parsimony, and out-of-sample performance. The final model achieved a validation AUC of 0.891 and a Gini coefficient of 0.782. An XGBoost benchmark was also evaluated and achieved lower validation discrimination than the logistic regression model.

2. Data and Initial Analysis

2.1. Data

The dataset contains 5,960 observations of home-equity loans and 12 explanatory variables describing loan characteristics, borrower employment and credit history. The target variable, **BAD**, indicates whether the applicant defaulted on the loan or became seriously delinquent (**BAD** = 1) or successfully repaid the loan (**BAD** = 0).

Variable	Description	Type
BAD	Default / serious delinquency indicator	Binary
LOAN	Loan amount requested	Continuous
MORTDUE	Existing mortgage amount due	Continuous
VALUE	Current property value	Continuous
REASON	Loan purpose	Categorical
JOB	Occupational category	Categorical
YOJ	Years at current job	Continuous
DEROG	Major derogatory reports	Count
DELINQ	Delinquent credit lines	Count
CLAGE	Age of oldest credit line	Continuous

NINQ	Recent credit inquiries	Count
CLNO	Number of credit lines	Count
DEBTINC	Debt-to-income ratio	Continuous

2.2. Train/Test Split

The dataset was divided into a 70% training sample (4,172 observations) and a 30% validation sample (1,788 observations) using stratified sampling based on the target variable. All variable transformation, binning and model development decisions were performed using the training sample, while the validation sample was reserved for final out-of-sample performance assessment.

3. Variable Analysis and Preparation

Missing values were present in several variables. Their relationship with the target variable was assessed before deciding on the treatment. In particular, missing values in VALUE and DEBTINC showed substantially different default rates from their observed counterparts and were therefore retained as separate categories during modelling. Other variables were handled using training-sample-based imputation where appropriate.

Variable	Finding / issue	Treatment
LOAN	No NaNs; Right-skewed; nonlinear relationship with log-odds	Log transformation, $\log(\text{LOAN} + c)$, where $c = 550$
MORTDUE	Right-skewed; nonlinear relationship with log-odds	Mean imputation; Log transformation, $\log(\text{MORTDUE} + c)$, where $c = 1031.5$; Removed from final model
VALUE	Strongly different default rate for missing values; Right-skewed; nonlinear relationship with log-odds	Separate category for NaNs; Binned: <50k, >50k, Missing
REASON	No significant discriminant differences among categories	Mode imputation; Removed from final model
JOB	Different default rates across categories; sparse categories	Separate category for NaNs; Categories Merged;
YOJ	Nonlinear relationship with default	Binned: 0–5, 5–20, 20+, Missing
DEROG	Distinct concentration at zero; extreme outliers; sparse right tail	Binned: 0, 1–2, 3+, Missing
DELINQ	Distinct concentration at zero; extreme outliers; sparse right tail	Binned: 0, 1–4, 5+, Missing
CLAGE	Continuous predictor with	Mean imputation; Log

	symetrical distribution with some right tail outliers	transformation, $\log(\text{CLAGE} + c)$, where $c = 0.243$
NINQ	Distinct concentration at zero; extreme outliers; sparse right tail	Binned: 0, 1-3, 3+, Missing
CLNO	Continous predictor with symetrical distribution with some right tail outliers	Mean imputation; Log transformation, $\log(\text{CLNO} + c)$, where $c = 0.5$
DEBTINC	Strongly different default rate for missing values; dense around mean with sparse tails	Binned: <40, 40+, Missing

Based on the univariate analysis and data preparation, the transformed variables were used to construct the adjusted logistic regression model. Variables that did not provide significant incremental explanatory power were subsequently considered for removal during multivariate model development.

4. Model Development

4.1. Baseline Logistic Regression

As a benchmark, a baseline logistic regression model was developed using the original variables. Missing continuous values were replaced using training-sample means, while missing categorical values were replaced using training-sample modes. Categorical variables were one-hot encoded. No transformations or binning were applied at this stage.

The baseline model contained 17 estimated parameters and achieved a training AUC of 0.807 and a validation AUC of 0.772, corresponding to a validation Gini coefficient of 0.544. This model provides a reference point for assessing the impact of the subsequent variable transformations and binning.

4.2. Adjusted Logistic Regression

The adjusted model incorporated the transformations and binning identified during the variable analysis. Continuous variables were log-transformed where appropriate, while variables exhibiting nonlinear relationships were grouped into bins. Missing values with informative differences in default rates were retained as separate categories.

The adjusted model contained 26 estimated parameters and achieved a training AUC of 0.911 and a validation AUC of 0.891, corresponding to a validation Gini coefficient of 0.781. This represented a substantial improvement in discriminatory performance compared with the baseline model.

4.3. Variable Reduction

The adjusted model was subsequently examined for variables that did not provide significant incremental explanatory power after controlling for the remaining predictors. MORTDUE and

REASON were not statistically significant in the multivariate model and were therefore removed from the reduced specification.

The reduced and full adjusted models were compared using a likelihood-ratio test. The test resulted in a statistic of 0.189 with 2 degrees of freedom and a p-value of 0.910, providing no evidence that retaining the two variables significantly improved model fit. The reduced model therefore provided a more parsimonious specification while maintaining comparable predictive performance.

The reduced model contained 24 estimated parameters and achieved a training AUC of 0.912 and a validation AUC of 0.891, corresponding to a validation Gini coefficient of 0.782.

4.4. Alternative Model — XGBoost

As an alternative modelling approach, an XGBoost classifier was trained using the same prepared feature set. It achieved a validation AUC of 0.788 and a Gini coefficient of 0.576. Although XGBoost achieved higher threshold-based classification metrics at a 0.5 threshold, its overall ranking performance was lower than that of the logistic regression models.

Given the objective of developing an interpretable PD scoring function, the logistic regression approach was retained.

5. Final Model and Validation

Model	Train AUC	Validation AUC	Validation Gini
Baseline Logistic	0.807	0.772	0.544
Adjusted Logistic	0.911	0.891	0.781
Reduced Logistic	0.912	0.891	0.782
XGBoost	0.845	0.788	0.576

The reduced logistic regression model was selected as the final specification based on statistical significance, model parsimony, and validation performance. It achieved a training AUC of 0.912 and a validation AUC of 0.891, corresponding to a validation Gini coefficient of 0.782.

The small difference between training and validation AUC indicates stable discriminatory performance on the held-out sample. Further validation, including out-of-time testing and calibration assessment, would be recommended before production deployment.